Jeevan Sen Joseph

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PROFESSIONAL SUMMARY

AI/ML Engineer who works directly with clients to scope ambiguous problems and ship production-ready AI systems end to end. Distilled Amazon Nova Pro into Nova Lite on AWS Bedrock (78% → 95% extraction accuracy), took a natural language-to-query system (Django, PostgreSQL, MongoDB) from prototype to a scalable product on live databases, and served as a client-facing AI advisor translating technical tradeoffs for non-technical stakeholders. Hands-on across LLM fine-tuning and distillation (LoRA, IA3, PyTorch, CUDA), RAG (FAISS, embeddings), and data pipelines.

EDUCATION

APJ Kalam Technological University *Nov 2021 – Jun 2025*
Bachelor of Technology (B.Tech.) in Artificial Intelligence and Data Science

TECHNICAL SKILLS

- **Programming Languages:** Python (proficient), SQL (proficient); Java, C++ (familiar)
- **Data Structures & Algorithms:** Trees, Graphs, Hash Maps, Lists, Sorting/Searching — applied in efficient pipeline and agent design
- **Algorithm Design & Optimization:** ML Algorithms, Search & Retrieval, Feature Engineering, RAG Optimization, LLM Inference & Fine-Tuning
- **Version Control:** Git, GitHub
- **Cloud & Managed AI Services:** AWS Bedrock (model distillation workflow), AWS EC2, Amazon Nova Pro, Amazon Nova Lite
- **Application Development:** Django, Full-Stack Development, REST API Development, Natural Language to SQL (NL2SQL), Schema-Aware Query Generation
- **RAG & Vector Search:** Retrieval-Augmented Generation (RAG), FAISS, Vector Databases, Embedding Models, Semantic Search
- **Data & Databases:** Pandas, NumPy, Data Cleaning, Data Pipelines, PostgreSQL, MySQL, MongoDB, NoSQL, SQL
- **Machine Learning & LLM Engineering:** PyTorch, TensorFlow, Scikit-learn, Deep Learning, CUDA / GPU Computing, LLM Fine-Tuning (LoRA, IA3, PEFT), Knowledge Distillation, Model Compression, Mixture of Experts (MoE)
- **LLM Platforms & Models:** GPT (OpenAI), LLaMA, Mistral, HuggingFace (Transformers, Datasets & Tokenizers), Accelerate, Prompt Engineering (One-Shot, Few-Shot)
- **Agentic AI:** LangGraph, Multi-Agent Systems, Supervisor-Agent Orchestration, Function Calling / Tool Use, Model Context Protocol (MCP)
- **Process & Collaboration:** Software Development Life Cycle (SDLC), Agile/Scrum, Cross-Functional Collaboration, Client Collaboration
- **Certifications:** NLP, Deep Learning, AI, OpenAI GPT Models, Data Science (Infosys Springboard)

EXPERIENCE

IBS Software

Aug 2024 – Nov 2025

AI Research Engineer *May 2025 – Nov 2025*

- Tackled an email information-extraction project (~21 fields/email) with two distillation approaches. On AWS EC2, with direct access to model weights, engineered a custom logit-based pipeline (Python, PyTorch, CUDA) compressing LLaMA 7B → 3B, also evaluating Mistral as a candidate model — teacher–student forward pass alignment, logit extraction, masking, and loss computation — achieving 15% accuracy gains over baseline and 12% improvement on the compressed student model.
- On AWS Bedrock, distilled Amazon Nova Pro into Amazon Nova Lite through the managed distillation workflow for the same project, lifting field-level accuracy from 78% to 95%.
- Led production engineering from a base LLM to a domain-specific, production-ready LLM.
- Scaled training with HuggingFace Accelerate (distributed training, gradient accumulation) for computational efficiency, achieving significant inference speed-up while preserving domain accuracy; defined evaluation metrics (perplexity, logit similarity, domain-match score).
- Collaborated cross-functionally with the data team to source distillation data; ran large-scale data analytics, cleaning, and pipelining, building preprocessing pipelines with TensorFlow and HF Datasets & Tokenizers.
- Solved ambiguous AI engineering problems by evaluating competing approaches across accuracy, performance, and implementation complexity.
- Applied MoE and IA3 adaptation to improve computational efficiency and cut the cost of deploying domain-specific models into production.
- Engineered a reusable fine-tuning pipeline (Python, PyTorch, HuggingFace, PEFT) supporting LoRA and IA3, cutting compute cost while achieving up to 13% accuracy improvement (LoRA + IA3 combined) for lower-cost deployment of adapted models.
- Built the full data cleaning, feature engineering, and preprocessing pipeline for text classification, from raw data to tokenized training-ready inputs.

AI/ML Engineer *Aug 2024 – Mar 2025*

- Built and shipped a full-stack, production-facing ML application (Python, Django, REST APIs, GPT-3.5 Turbo) converting natural language into schema-aware SQL (PostgreSQL) and NoSQL (MongoDB) queries, taking it through production engineering from prototype to a scalable product.
- Used prompt engineering with one-shot and few-shot prompting to guide GPT-3.5 Turbo toward accurate, schema-aware SQL and NoSQL query generation.
- Applied Agile methodology and software development practices and standards across the full SDLC to deliver robust, maintainable, scalable software, with a database-agnostic plugin query layer that runs on MongoDB and PostgreSQL and extends to MySQL and other databases with minimal changes.
- Leveraged AI-assisted development tools to accelerate implementation, debugging, refactoring, and test development, independently validating generated code to maintain production-quality engineering standards.
- Implemented input validation, admin-level execution controls, query context checks, and ethical AI / bias mitigation in SQL generation — engineering safeguards for a system running against live databases.

ArkytUp

Freelance AI Consultant *Jul 2025 – Present*

Client Engagement (confidential AI product)

- Served as sole, client-facing AI advisor to an early-stage startup, driving AI technology selection and de-risking technical feasibility.
- Ran a comparative evaluation of candidate ML approaches for transforming one data type into another — weighing accuracy, performance, and implementation complexity — and delivered research-backed model and tooling recommendations.
- Translated complex technical tradeoffs into clear, actionable recommendations for non-technical stakeholders.
- Shaped system design and the development roadmap through close client collaboration and recurring technical reviews with the founder.

Independent Prototypes (separate from the client product)

- PDF-to-Speech RAG: Built a conversational AI system (Python, FAISS, OpenAI embeddings + API, speech I/O) for semantic search, summarization, and Q&A over PDFs; grounded responses through chunking, embeddings, and similarity-based retrieval to reduce hallucination, guided by RAG evaluation metrics.
- Multi-Agent Trip Planner: Built a LangGraph multi-agent system where a supervisor routes tasks to specialized agents using function calling and MCP tools.

Wytty Platforms

Product Engineer *Jan 2023 – Feb 2024*

- Co-created Wytty, a social platform organizing interaction around shared interests rather than followers, helping users discover their "tribe" of like-minded people.
- Drove product vision and roadmap, enabling users to join interest-based communities, share content, and build connections around what they genuinely care about.